Conservative Road-Scene Risk Assessment via Fail-Aware Multi-Model Fusion

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***Abstract*—Single-image assessment of road hazards is limited by detector misses, scale-ambiguous monocular depth, and uncertain object-to-path relationships. This paper presents a fail-aware fusion layer that combines YOLO11 object evidence, HybridNets drivable-area and lane cues, and Depth Anything V2 relative depth while keeping component uncertainty explicit until the advisory stage. Scene-quality assessment, a multi-region object-depth consistency check, path-relation reasoning, severity gates, and a depth-only corridor fallback turn the evidence into a conservative advisory with an audit record for every object. The frozen pipeline is evaluated against BDD100K annotations on a 9,500-image validation test split and a reproducible 100-image hard set. Of the accepted test detections, 84.9% receive a STABLE depth estimate, and non-STABLE estimates are 1.8 times more likely to be false positives. Unsafe CLEAR advisories occur on 11 of 3,250 images whose ego lane holds a relevant road user. This safety comes at a measured cost: CLEAR is issued on only 2.1% of test images, mainly because a perspective ego corridor assigns many roadside and distant objects to the path. Cached-replay ablations, an operating curve, and perturbation tests quantify each component. Code and image-ID manifests are released.**

***Index Terms*—advanced driver assistance systems, BDD100K, monocular depth estimation, multi-model fusion, selective prediction, uncertainty, YOLO11.**

I. INTRODUCTION

Driver-assistance perception commonly combines object detection, drivable-area and lane segmentation, and monocular depth estimation, each with its own failure modes. A detector can localize a vehicle without determining whether it lies in the ego path; a drivable-area model can mark road pixels as traversable although an obstacle occupies them; and monocular depth orders a scene from near to far, but its scale is ambiguous and its local estimates become unstable under darkness, glare, blur, weak texture, or occlusion [1], [2].

The practical challenge is therefore to decide when the combined evidence is strong enough to support an advisory. Collapsing detector, road, lane, and depth outputs into one precise-looking warning hides uncertainty, especially for a single image, where ego speed, object velocity, and temporal consistency are unavailable. A conservative system should avoid unsupported clear-path conclusions and expose uncertainty when evidence is weak or inconsistent, in the spirit of selective prediction [3] and run-time monitoring of learned perception [4].

This paper studies conservative road-hazard assessment from a single front-facing image. Object, road, depth, and image-quality evidence stay separate until the advisory stage: YOLO11 provides detections and instance masks, HybridNets drivable-area and lane cues, and Depth Anything V2 a relative-depth field [2], [5], [6]. Weak or incomplete evidence may raise caution, for example from CLEAR to MONITOR, but can never create object evidence or verify a driving action. The contributions are:

- an interpretable fusion architecture that relates YOLO11 object evidence to a HybridNets-based ego corridor and Depth Anything V2 relative depth, in which weak evidence can add caution but cannot justify a severe advisory;
- a multi-region object-depth reliability estimator that reports STABLE, CHECK, or UNCERTAIN, whose states are validated against BDD100K annotations;
- severity gates and a depth-only corridor fallback, measured with ground-truth ego-lane occupancy, and an operating curve that makes the cost of conservatism explicit;
- a reproducible evaluation on 9,500 BDD100K validation images and a deterministic hard set, with released image IDs, cached-replay ablations, and image-level confidence intervals.

II. RELATED WORK

Road-scene perception. BDD100K covers diverse weather, lighting, and road conditions [1]. YOLO-family detectors are widely used for traffic objects [6], [7], but a box does not say whether an object occupies the ego path; multi-task networks such as YOLOP and HybridNets add drivable-area and lane context [5], [8].

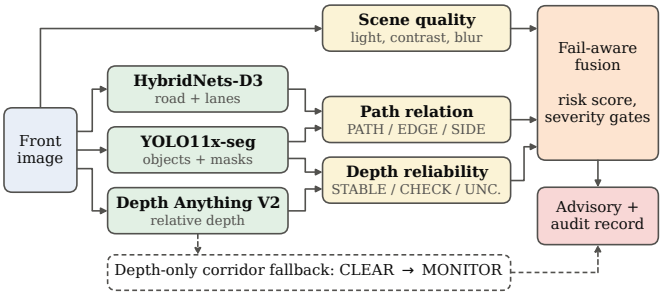


Fig. 1. Fail-aware fusion pipeline. Component outputs remain explicit until reliability and geometry gating; the depth-only fallback can only raise CLEAR to MONITOR.

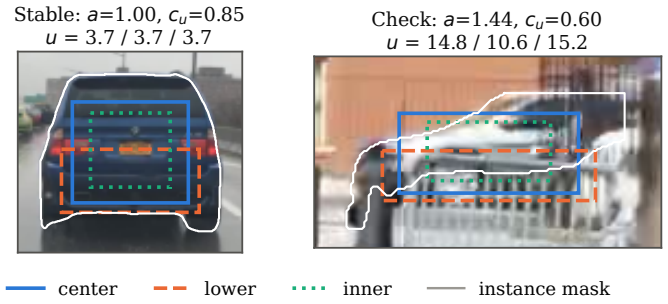


Fig. 2. Multi-region depth sampling. Left: the three regions of a vehicle agree, so the estimate is STABLE. Right: the mask leaks into the background, the regions disagree, and the estimate is kept at CHECK.

Depth and uncertainty. MiDaS and Depth Anything predict affine-invariant relative depth that orders a scene well but lacks metric scale [2], [9], [10]. Neural confidence can be poorly calibrated [11]; Bayesian modeling, deep ensembles, and depth-specific estimators address uncertainty inside the network [12], [13], [14], [15]. This work instead operates at the fusion level, on frozen models, and expresses uncertainty through agreement between independent evidence sources.

Selective prediction and monitoring. Selective classification abstains when confidence is insufficient and is judged by its risk–coverage trade-off [3]; conformal prediction provides distribution-free uncertainty sets [16]; run-time monitors flag unreliable perception outputs [4]; and probabilistic detection [17] and adverse-weather studies [18] motivate fail-aware perception. Our pipeline applies these ideas to multi-model static-image fusion and reports the trade-off between unsafe CLEAR outputs and over-caution.

III. FAIL-AWARE FUSION METHOD

A. Complementary Evidence

The system combines three pretrained perception sources (Fig. 1): YOLO11 object classes, confidences, boxes, and instance masks [6]; HybridNets drivable-area and lane masks, used to estimate an internal ego corridor [5]; and a Depth Anything V2 relative-depth field [2]. No component is trained or fine-tuned for this work. Object evidence comes from the detector only: road and depth cues can change an advisory, but they never create an object or verify a driving action. Depth is used only as a relative cue, never as metric distance, speed, or time-to-collision. Uncertainty is handled asymmetrically: weak evidence lowers the confidences below and can add caution (the fallback turns CLEAR into MONITOR), but the gates of Sec. III-C keep it from justifying BRAKE or STOP.

B. Scene and Object-Depth Reliability

Scene quality is estimated from the gray image: median illumination m, the 10–90 percentile range R, and the Laplacian variance v. With $q_{\text{illum}} = \text{clip}((m - 18)/72)$, $q_{\text{contrast}} = \text{clip}((R - 20)/80)$, and q_{sharp} the log-scaled variance between 12 and 500, all clipped to $[0, 1]$, $q_s = 0.40\, q_{\text{illum}} + 0.30\, q_{\text{contrast}} + 0.30\, q_{\text{sharp}}$. Low light ($m < 45$), low contrast ($R < 55$), blur ($v < 60$), and glare are kept as separate flags; a scene is DEGRADED when $q_s < 0.35$ or it is both dark and blurred, and LIMITED when $q_s < 0.58$ or any flag is raised.

Depth Anything V2 outputs an affine-invariant inverse-depth response d in which larger values are nearer [2], [9]. The pipeline converts it into a relative depth index $u = \text{clip}(1000/d,\, 0.1,\, 999)$, so that a *smaller* u means a *nearer* surface. Because d has an unknown scale and shift, u preserves only the within-image ordering; it is not a distance and is never compared across images.

For each accepted object, u is sampled in three regions of the eroded instance mask: a center region, a lower-body region, and an inner-core region (Fig. 2). Let u_r be the trimmed (12–88 percentile) median in region r and n_r its number of retained pixels. The object estimate is the support-weighted geometric mean $u_o = \exp(\sum_r w_r \log u_r / \sum_r w_r)$ with $w_r = \min(n_r, 300)$, and regional consistency is measured by the disagreement ratio $a = \max_r u_r / \min_r u_r$. With s the 20–80 interquantile spread of all sampled values normalized by u_o and N the number of retained pixels, the base reliability is

$$c_u = \text{clip}(1.35/a, 0, 1)\, e^{-0.85s}\, \text{clip}(N/120,\, 0.25,\, 1), \quad (1)$$

which is then multiplied by a mask-quality factor $\text{clip}(0.45 + 0.75\, m_q,\, 0.35,\, 1)$ (where m_q penalizes nearly empty or box-like masks), by 0.58 when only a box is available, and by 0.72 or 0.82 for objects truncated by the side/top or bottom image border. The estimate is STABLE when $c_u \geq 0.68$ and $a \leq 1.35$, CHECK when $c_u \geq 0.35$ and $a \leq 1.80$, and UNCERTAIN otherwise. The constants were fixed during development, before the evaluation reported here; Section V-F reports their sensitivity.

C. Path Relation and Conservative Advisory

The ego corridor is a perspective band that starts at 35% of the image height and widens toward the bottom; its center and width follow the HybridNets lane pixels in the rows where both lane boundaries are visible and keep the default trapezoid elsewhere. An object is PATH when its ground-contact point lies inside the corridor or at least 10% of the lower 30% of its

mask overlaps it, EDGE when it overlaps a dilated corridor, and SIDE otherwise.

A risk score combines the relative depth index, image area, bottom position, path relation (+32 for PATH, +15 for EDGE), and a vulnerable-road-user bonus, and is mapped to CLEAR, MONITOR, SLOW, BRAKE, or STOP at the cut-offs 24, 40, 60, and 78. The evidence confidence combines detector confidence c_{det}, depth reliability c_u, a distance-agreement score c_d, and scene quality:

$$c_e = \sqrt{c_{\mathrm{det}}\ \max(c_u, c_d)}\ (0.65 + 0.35\, q_s). \tag{2}$$

Here c_d measures the agreement of up to three monocular distance cues (depth fitted to the road plane, ground-contact geometry, and class-height priors) under default camera assumptions (1.5 m height, 70° field of view). It is used only as an agreement score inside (2) and the weak-evidence gates; no metric distance is reported.

Severity gates then cap unsupported outputs and can only lower an advisory: (i) a SIDE object raises at most MONITOR, or SLOW for a close vulnerable road user at the image bottom; (ii) an EDGE object raises at most BRAKE; (iii) STOP requires a STABLE, very near PATH object ($u_o \le 4$ with $c_u \ge 0.60$, at least 2.5% of the image area, and its lower edge in the bottom 32% of the image, or a distance cue of at most 4 m with $c_d \ge 0.55$) with $c_{\mathrm{det}} \ge 0.55$ and $c_e \ge 0.48$; (iv) BRAKE requires a credible size or position, $c_{\mathrm{det}} \ge 0.30$, and $c_e \ge 0.28$; (v) an UNCERTAIN object with $c_{\mathrm{det}} < 0.35$ raises at most MONITOR; and (vi) in a degraded scene ($q_s < 0.35$), BRAKE also requires a STABLE object with $c_{\mathrm{det}} \ge 0.45$. A failed gate lowers the advisory by one level.

D. Depth-Only Corridor Fallback and Audit Record

Detector-empty images should not automatically be treated as safe. The corridor is divided into far, middle, and near bands, and the median u of road pixels in each band is checked for the expected monotonic order. The corridor state is UNAVAILABLE when fewer than 55% of the band pixels are valid or a band is missing, LIMITED when the scene is degraded or the band confidence is below 0.38, CHECK when the ordering is violated, and CONSISTENT otherwise. When no object is accepted and the state is not CONSISTENT, the global advisory is raised from CLEAR to MONITOR. The fallback never creates an object, a location, or a distance.

The output is the global advisory plus one audit record per accepted object (class, confidences, mask source, u_o, reliability state, path relation, risk score, and advisory), so every advisory can be traced to its evidence. A STABLE state indicates regional agreement, not ground-truth correctness.

IV. EXPERIMENTAL PROTOCOL

A. Models and Inference Profile

YOLO11x-seg (COCO-pretrained) runs at 960 px with a confidence threshold of 0.15 and four test-time passes (raw and scene-adaptively enhanced input, each with a horizontal flip) fused by weighted box fusion; single-pass detections below 0.34 are rejected. Enhancement (gamma, luma CLAHE, mild unsharp masking) is applied to the YOLO input only, when the scene flags call for it. HybridNets-D3 runs at 640 px and Depth Anything V2 ViT-L at 518 px. All thresholds and fusion rules are fixed across every evaluation; the code is commit `4a5051e`, and the weight checksums are listed in the repository.

B. Evaluation Data

All headline experiments use the 10,000 BDD100K validation images [1], whose labels none of the three networks was trained on. A random 500-image *dev* subset (seed 20261108) is used only for threshold sensitivity, the operating curve, and the dataset-average priors of the ablations. The remaining 9,500 images form the *test* split on which every headline number is computed (ID-list SHA-256 prefix `af51f530c5d9b3c5`). The *hard* set is a deterministic stress subset of the test split: all validation images are measured at 320×180 (mean brightness b, standard deviation c, Laplacian variance l, dark and bright pixel fractions f_d (≤ 35) and f_b (≥ 245), Canny-edge density e), each normalized between its 2nd and 98th percentiles; the category scores are $0.65 f_d + 0.35(1-b)$ (dark), $1-c$ (low contrast), $1-l$ (blur), $0.75 f_b + 0.25 b$ (glare), and e (complex), and each category in turn keeps its 20 highest-scoring images not already selected (prefix `5d6c11f5bdd1198a`). Perturbation stability is also measured on a random 1,000-image subset of the test split (the *1k subset*), and the 8,400 test images outside the hard set and the 1k subset (*val_rest*) serve as a held-out check of the operating point chosen on the dev subset. As a supplementary check, the same protocol is applied to the 69,863 annotated BDD100K *train* images (labels from public copies that reproduce all 10,000 validation labels and masks exactly); HybridNets was trained on them, so they are reported separately.

C. Ground-Truth Protocol

BDD100K validation annotations provide boxes with occlusion and truncation flags, time-of-day and weather tags, and drivable-area maps that separate the *direct* (ego) lane from *alternative* lanes [1]. A *road user* is any annotated person, rider, or vehicle; it is *relevant* when its box is at least 5% of the image height (36 px, roughly within 40 m for a car). Its ground-truth lane is the drivable label under a ground-contact strip just below the box (direct or alternative when those labels cover at least a quarter of the strip). Detections are matched class-agnostically to road users at IoU ≥ 0.5. An image is *occupied* when a relevant road user stands in the direct lane. An *unsafe* CLEAR is a CLEAR advisory on an occupied image; *over-caution* is SLOW or higher when no road user of any size is in the direct lane (an upper bound: an adjacent-lane hazard can justify such advisories); and a BRAKE/STOP advisory is *unsupported* when none of the objects that raised it is a true positive in the direct or alternative lane. These definitions were fixed before any result was inspected.

D. Ablations, Replays, and Statistics

Each ablation re-runs only the fusion stage on identical network outputs, so all comparisons are paired; replaying the

TABLE I
INTERNAL RELIABILITY AND GROUND-TRUTH RESULTS

Measure	Hard (100)	Test (9,500)
Accepted detections	505	75,819
STABLE/CHECK/UNCERTAIN	437/67/1	64,334/11,104/354
Stable ratio, Wilson 95% CI	86.5 [83.3, 89.2]	84.9 [84.6, 85.1]
Stable ratio, image-bootstrap CI	86.5 [83.5, 89.2]	84.9 [84.6, 85.1]
Detection precision (GT)	86.3 [82.9, 89.6]	89.1 [88.8, 89.4]
Recall, relevant road users	72.2 [67.2, 77.3]	78.9 [78.4, 79.3]
PATH precision (matched)	15.8 [11.0, 22.3]	9.2 [8.8, 9.5]
PATH recall (matched)	56.8 [42.2, 70.3]	53.8 [52.3, 55.2]
Corridor IoU with GT ego lane	0.242	0.240
Occupied images	30	3,250
Unsafe CLEAR, fallback on	0/30 (0.0)	11/3,250 (0.3)
Unsafe CLEAR, fallback off	1/30 (3.3)	15/3,250 (0.5)
Over-caution (free ego lane)	34/67 (50.7)	4,799/5,844 (82.1)
Unsupported BRAKE/STOP	31/47	4,179/6,188
Detector-empty / fallback fired	15/15	295/236

TABLE II
RELIABILITY STATES AGAINST BDD100K ANNOTATIONS (TEST SET, %)

State	n	False positive	Occl.	Trunc.	IoU
STABLE	64,334	9.7 [9.5, 10.0]	59.0	2.5	0.825
CHECK	11,104	16.5 [15.8, 17.2]	35.5	69.2	0.849
UNCERTAIN	354	44.1 [38.9, 49.3]	54.5	7.1	0.716

published constants reproduces the original reports byte for byte. The cumulative baselines are B0, YOLO with dataset-average road and depth priors from the dev subset; B1, which adds the HybridNets corridor; B2, which adds Depth Anything V2 sampled in the center region only; and B3, which adds the multi-region reliability and scene quality. The full system adds the severity gates and the fallback. Stability is measured under six mild perturbations (darker and brighter gamma, 0.7 contrast, Gaussian blur with $\sigma = 1.5$, Gaussian noise with $\sigma = 8$, and JPEG quality 35). Proportions use Wilson intervals [19]; object-level ratios use an image-level cluster bootstrap with 10,000 resamples [20], because detections in one image are not independent; and paired image-level outcomes use the exact McNemar test [21]. Inference for 3,270 test images and every perturbation run of the 1k subset ran on an NVIDIA T4 with the same pinned library versions; perturbed images are always compared with clean runs from the same GPU.

V. RESULTS

A. Internal Reliability and Detection Correctness

Table I summarizes the test split and the hard set. On the test split, 64,334 of 75,819 accepted detections are STABLE (84.9%, image-bootstrap CI 84.6–85.1%); on the hard set the ratio is 86.5% (Wilson CI 83.3–89.2%), close to the 86.1% of the submitted version.[1] Against the annotations, the accepted test detections reach 89.1% precision and 78.9% recall of relevant road users; relative to the raw YOLO ensemble (85.3% precision, 82.2% recall), the fusion-level suppression trades a little recall for precision.

[1]The submitted version reported 596 detections on a hard set whose image list was not preserved; every number here comes from the regenerated, published splits.

TABLE III
CUMULATIVE BUILD-UP ON THE TEST SET (%). FLIP: ADVISORY CHANGE UNDER MILD PERTURBATION (HARD SET)

Configuration	Unsafe	%	Over-c.	Severe	Flip
B0: YOLO only (road/depth priors)	17/3,250	0.5	86.7	66.8	13.0
B1: + HybridNets corridor	16/3,250	0.5	88.4	65.3	–
B2: + DA V2, single region	15/3,250	0.5	88.1	74.3	–
B3: + multi-region, scene quality	15/3,250	0.5	88.2	74.5	14.5
Full: + gates, fallback (proposed)	11/3,250	0.3	82.1	65.1	8.3

B. Do Reliability States Mean Anything?

Table II relates the reliability states to the annotations. Non-STABLE test estimates are false positives in 17.4% of cases (CI 16.7–18.1%) against 9.7% for STABLE ones (CI 9.5–10.0%), a factor of 1.8. CHECK concentrates on objects truncated by the image border (69.2% vs. 2.5%), exactly the case where regional depth samples cannot agree. The states are thus informative about correctness, although STABLE remains an agreement measure, not a depth-accuracy guarantee.

C. Advisories, Path Relation, and the Cost of Conservatism

Unsafe CLEAR occurs on 11 of 3,250 occupied test images (Wilson upper bound 0.6%) and on 0 of 30 occupied hard-set images. In 6 of the 11 test cases the in-lane road user was detected, but the corridor classified it as SIDE, which caps its advisory; in the other 5 the detector missed it (4 at night); the 61 train cases split the same way (30 and 31). This has a cost: only 2.1% of test images receive CLEAR and 65.1% receive BRAKE or STOP, over-caution affects 4,799 of 5,844 free-lane images, and 67.5% of BRAKE/STOP advisories are unsupported by an in-lane object. The main cause is the ego corridor. Matched PATH precision is only 9.2% (recall 53.8%) because the perspective band is wider than the true ego lane near the vanishing point, so roadside and distant road users are labeled PATH (23,007 of the 25,730 matched PATH objects stand outside the annotated drivable area). The corridor overlaps the annotated ego lane with IoU 0.24, although the HybridNets road mask itself matches the annotated drivable area with IoU 0.83. Part of the disagreement is annotation: on unmarked streets the BDD100K direct lane often spans the whole carriageway. Fig. 3 shows both behaviors, and Fig. 4 places the published cut-offs on the trade-off between unsafe CLEAR and over-caution. On the dev subset, the cut-offs can be relaxed up to $\times 1.4$ before the first unsafe CLEAR, which lowers over-caution from 85.0% to 72.4% (at $\times 2.5$ it falls to 11.6% while 13.0% of occupied images receive CLEAR); on the 8,400 held-out val_rest images the same relaxation raises unsafe CLEAR from 9 to 28 of 2,867 occupied images (0.31% to 0.98%, upper bound 1.41%) while over-caution falls from 82.0% to 70.2%, and on the 69,863 train images unsafe CLEAR moves from 0.26% to 0.77% and over-caution from 80.8% to 69.9%. A margin chosen on the 500 dev images therefore does not transfer, and the published cut-offs are kept.

D. Ablations

Table III builds the system up from YOLO alone on the test split, and leave-one-out replays isolate each component

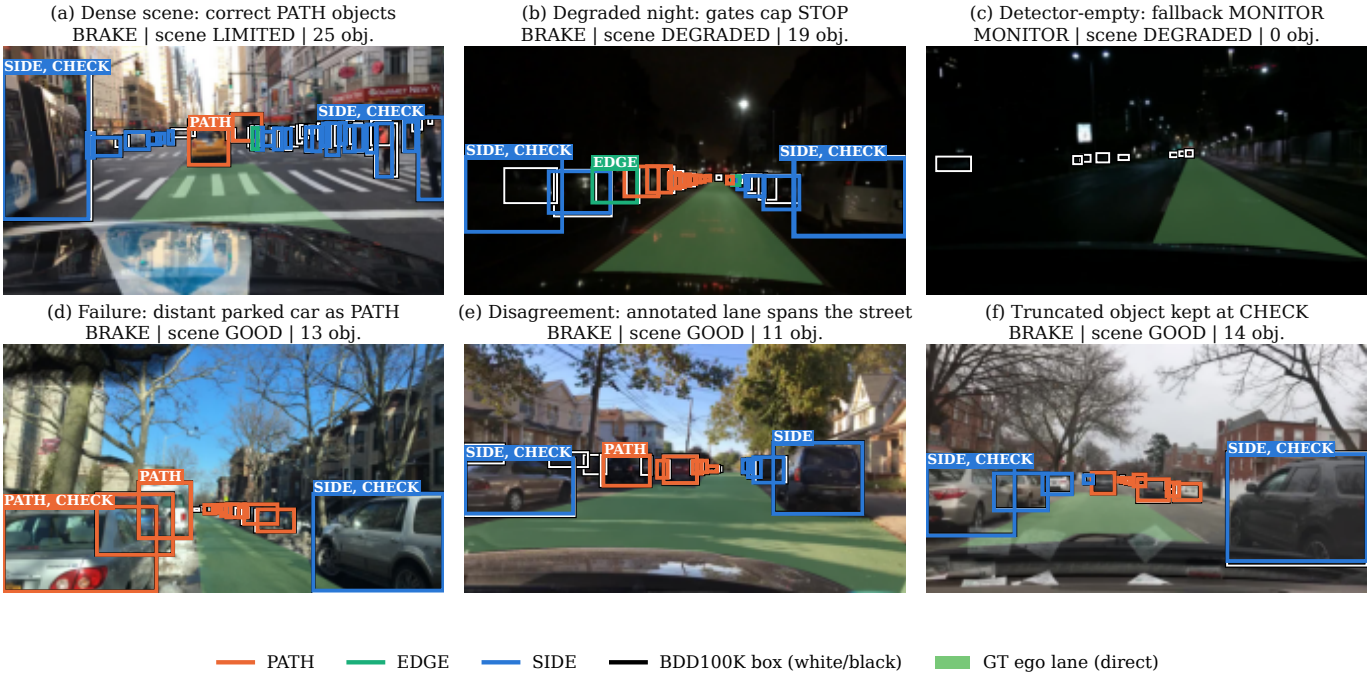


Fig. 3. Hard-set cases selected by fixed rules: three intended behaviors (a–c) and three failure modes or annotation disagreements (d–f). Colored boxes are system outputs labeled by path relation (CHECK marks a depth estimate that is not STABLE; small boxes are unlabeled); white boxes are BDD100K annotations; the green tint is the annotated ego lane.

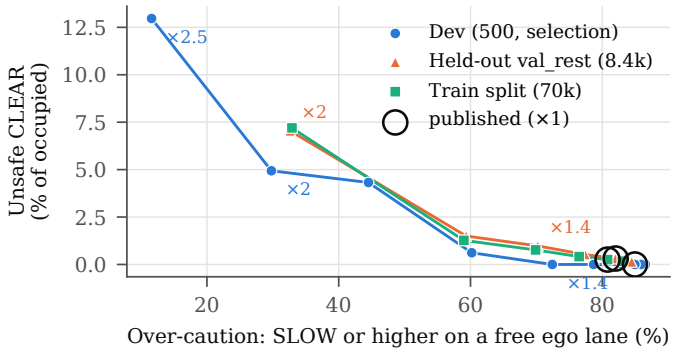


Fig. 4. Operating curve: all advisory cut-offs scaled together (dev subset: $\times 0.5$ to $\times 2.5$; held-out val_rest and train split: $\times 0.8$ to $\times 2.0$; circled points: published constants).

TABLE IV
RESULTS BY DIFFICULTY CATEGORY, TIME OF DAY, AND WEATHER (%)

Group	Img.	Obj.	Stable	Prec.	Rec.	Unsafe CLEAR
Hard set, by difficulty category						
Blur	20	43	81.4	83.7	56.9	0/3
Dark	20	52	90.4	82.7	66.1	0/2
Low contrast	20	93	81.7	72.0	69.0	0/7
Glare	20	110	89.1	93.6	78.4	0/5
Complex	20	207	87.4	90.3	77.2	0/13
Test split, by time of day and weather						
Daytime	4,993	46,431	86.6	92.9	83.3	7/1,944
Dawn/dusk	739	6,381	85.4	91.7	82.2	0/269
Night	3,734	22,872	81.1	80.7	70.4	4/1,027
Clear	5,056	36,396	83.8	86.3	76.3	6/1,563
Rainy	708	5,155	83.3	88.5	75.9	0/220
Snowy	737	5,896	84.1	89.5	78.6	2/269
Foggy	13	85	74.1	83.5	76.1	0/4

on identical network outputs. Against the YOLO-only baseline B0, the full system is less over-cautious (82.1% vs. 86.7%, $p < 0.001$) with no more unsafe CLEAR (11 vs. 17). The severity gates mainly control the aggressive outputs: removing them changes STOP advisories from 1,166 to 3,893, over-caution from 4,799 to 5,157 images (McNemar $p < 0.001$), and unsupported BRAKE/STOP outputs from 4,179 to 4,857. The fallback fired on 236 of the 295 detector-empty test images; without it, unsafe CLEAR changes from 11 to 15 ($p = 0.12$). Scene quality feeds the same mechanism: without it, the depth-only scan of degraded detector-empty images looks consistent, and CLEAR images change from 200 to 354. Multi-region sampling mainly changes the reliability states (stable ratio 80.3% with the center region alone vs. 84.9%) rather than the advisories (47 changed images).

The two dense sources contribute very differently. Replacing Depth Anything V2 with a dataset-average depth prior changes 3,631 test advisories and the stable ratio to 74.4%, so per-image depth drives both the reliability states and the severity. Replacing HybridNets with an average road prior changes only 364 advisories, because the corridor follows the lanes only where both boundaries are visible: on average only 9.8% of corridor rows are lane-guided, and 82% of test images use the default trapezoid for more than three quarters of the corridor. HybridNets is therefore under-used rather than unreliable (Section VI).

E. Difficult Conditions

Table IV reports the hard-set categories and the test split by time of day and weather. Relevant recall is 70.4% at night against 83.3% in daytime, while the stable ratio changes less (81.1% vs. 86.6%): local object-depth agreement can survive darkness, whereas detector recall and scene quality do not. HybridNets remains a reliable road source at night (road IoU 0.83 vs. 0.83). On the night images, replacing Depth Anything V2 changes 38.8% of the advisories (38.2% by day) and replacing HybridNets 5.5% (2.7%): depth matters equally in both conditions, the road cues about twice as much at night but still far less than depth. Weather is confounded with time of day (61% of the clear-weather test images are night images, against 39% of the rainy ones), so it is compared within each time of day: rain lowers relevant recall from 83.9% to 80.7% by day and from 70.7% to 66.3% at night (708 rainy images), whereas snow hardly changes it (82.3% and 71.0%; 737 images). Fog is too rare in BDD100K (13 test images) to support any conclusion. The train split repeats this pattern on 5,070 rainy and 5,549 snowy images (rain: 83.8% to 79.8% by day, 71.8% to 64.2% at night; snow: 83.2% and 73.5%), and fog remains too rare there (130 images). On the train images, on which HybridNets was trained (road IoU 0.85 against 0.83), unsafe CLEAR occurs on 61 of 23,604 occupied images and over-caution on 80.8% of free-lane images (94 and 79.1% with HybridNets replaced by the dev road prior).

F. Sensitivity, Stability, and Runtime

Changing each reliability constant by $\pm 20\%$ on the dev subset mainly moves the stable ratio itself (by up to 34.0 points, almost entirely through the STABLE threshold), whereas the advisories move much less: the BRAKE/STOP rate changes by at most 5.0 points (when the STABLE threshold is raised) and over-caution does not change. The advisory trade-off is governed by the risk cut-offs, which is why Fig. 4 sweeps them; no sensitivity setting produces an unsafe CLEAR on any of the 162 occupied dev images. Under the six mild perturbations, the advisory changes in 5.2% of image pairs on the 1k subset and 8.3% on the hard set, against 6.6% and 12.2% without gates; reliability states of persisting objects change in 3.0% of cases. With the center region alone the flips are essentially unchanged (5.1% and 8.7%) and the state changes rise slightly (3.4%): the multi-region check stabilizes the states, not the advisories. Hardware matters far less: rerunning the clean 1k subset on the T4 instead of the RTX 3050 changes none of its 1,000 advisories and 1 of the 5,130 val_rest advisories computed on both GPUs. On an RTX 3050 laptop GPU the median latency is 2.3 s per image (HybridNets 0.07 s, Depth Anything V2 1.58 s, four-pass YOLO11x 0.32 s; peak memory 2.6 GB) and 1.5 s on an NVIDIA T4 (119 images), so this configuration is an offline analysis tool, not a real-time module.

VI. DISCUSSION AND LIMITATIONS

The ground truth supports the fail-aware design where it matters for safety, and shows that the published operating point is conservative in the expensive direction, because a fixed perspective corridor over-assigns roadside and distant objects to the ego path and, in about half of the unsafe CLEAR cases, places a detected in-lane object beside it. The ablations point to a remedy: HybridNets segments the drivable area well, but the corridor rarely uses it. In an exploratory replay that is *not* part of the evaluated method, intersecting the corridor with the dilated HybridNets drivable mask changes matched PATH precision from 9.2% to 13.5% and recall from 53.8% to 42.3%, and over-caution from 4,799 to 3,843 images, with 11 unsafe CLEAR. A corridor that follows the drivable area therefore trades some path recall for less over-caution.

The study has further limitations. Static images cannot show ego speed, object velocity, time-to-collision, or temporal consistency. The ground-truth lane relation is approximated from drivable-area maps, the relevance threshold is a size proxy for distance, and severe weather is under-represented. Depth Anything was trained with unlabeled image collections that include BDD100K imagery [10], so the depth branch may be favored on this dataset. The four-pass configuration is far from real time. Future work should use the drivable mask in the corridor, calibrated distance targets, video-based temporal evaluation, and human rating of the advisories.

TABLE V
LEAVE-ONE-OUT ABLATIONS ON THE TEST SET (p: EXACT MCNEMAR TEST ON UNSAFE CLEAR VS. FULL)

Variant	Unsafe	Over-c.	Severe	Changed	p
Full system	11/3,250	82.1	65.1	–	–
w/o HybridNets (road prior)	13/3,250	80.3	65.6	364	0.62
w/o DA V2 (depth prior)	12/3,250	83.0	55.8	3,631	1.00
w/o multi-region (center)	11/3,250	82.0	65.0	47	1.00
w/o scene quality in fusion	14/3,250	82.4	65.7	232	0.25
w/o severity gates	11/3,250	88.2	74.5	4,024	1.00
w/o depth-only fallback	15/3,250	82.1	65.1	236	0.12

VII. CONCLUSION

This paper presented a fail-aware fusion layer for conservative road-hazard assessment from a single front-facing image, which combines YOLO11, HybridNets, and Depth Anything V2 evidence while preserving uncertainty through scene quality, multi-region depth reliability, path reasoning, severity gates, and a depth-only fallback. On 9,500 BDD100K validation images and a reproducible hard set, the reliability states align with annotated false positives and truncation, and unsafe CLEAR advisories are rare (11 of 3,250 occupied images); the measured price is frequent over-caution caused by an over-inclusive ego corridor. The released code, image-ID manifests, and cached-replay protocol allow every number to be regenerated: https://github.com/VuTien1311/roadsense-risk-assessment.

APPENDIX

SUPPLEMENTARY RESULTS

REFERENCES

[1] F. Yu *et al.*, "BDD100K: A diverse driving dataset for heterogeneous multitask learning," in *Proc. IEEE/CVF Conf. Computer Vision and*

TABLE VI

SUPPLEMENTARY CHECK ON THE BDD100K TRAIN SPLIT. HYBRIDNETS WAS TRAINED ON THESE IMAGES, SO ITS ROAD AND LANE OUTPUT (AND THE FULL-SYSTEM PATH MEASURES) ARE OPTIMISTIC THERE; NO-HN REPLACES HYBRIDNETS BY THE DEV ROAD PRIOR AND IS FREE OF THAT OVERLAP. RATES WITH WILSON 95% CIS.

Measure	Test full (9,500)	Test no-HN (9,500)	Train full (69,863)	Train no-HN (69,863)
Detection precision (GT)	89.1	91.3	88.9	91.2
Recall, relevant road users	78.9	76.0	79.4	76.6
PATH precision	9.2	9.9	9.1	9.8
PATH recall	53.8	54.4	53.3	54.0
Road IoU (HybridNets vs GT)	0.828	–	0.854	–
Ego-corridor IoU	0.240	–	0.237	–
Unsafe CLEAR (% occupied)	0.34 [0.19, 0.61]	0.40 [0.23, 0.68]	0.26 [0.20, 0.33]	0.40 [0.33, 0.49]
Over-caution (% free lane)	82.1 [81.1, 83.1]	80.3 [79.3, 81.3]	80.8 [80.4, 81.2]	79.1 [78.7, 79.5]
BRAKE/STOP (% images)	65.1	65.6	64.9	65.4

TABLE VII

ADVISORY STABILITY UNDER SIX MILD PHOTOMETRIC PERTURBATIONS (%): ADVISORY FLIPS, SEVERE FLIPS, AND RELIABILITY-STATE FLIPS OF PERSISTING OBJECTS

Variant	Hard flip	Hard sev.	1k flip	1k sev.	State
Full system	8.3	3.7	5.2	2.0	3.0
w/o severity gates	12.2	6.2	6.6	2.8	3.0
w/o multi-region	8.7	3.8	5.1	2.0	3.4
w/o scene quality	11.2	4.5	5.2	2.4	3.0
B3 (no gates, no fallback)	14.5	8.0	6.5	3.0	3.0
B0 (YOLO only)	13.0	8.5	6.8	3.2	3.9

TABLE VIII

THRESHOLD SENSITIVITY ON THE DEV SPLIT (%)

Setting (dev set)	Stable	Unsafe	Over-c.	Severe
Published constants	85.1	0/162	85.0	65.8
stable_c=0.544	96.4	0/162	85.0	65.8
stable_c=0.816	51.2	0/162	85.0	60.8
stable_a=1.08	80.7	0/162	85.0	65.4
stable_a=1.62	85.2	0/162	85.0	65.8
check_c=0.28	85.1	0/162	85.0	65.8
check_c=0.42	85.2	0/162	85.0	65.8
check_a=1.44	85.2	0/162	85.0	65.8
check_a=2.16	85.1	0/162	85.0	65.8
k_ratio=1.08	84.3	0/162	85.0	65.8
k_ratio=1.62	85.1	0/162	85.0	65.8
k_spread=0.68	86.2	0/162	85.0	65.8
k_spread=1.02	83.9	0/162	85.0	65.8
n_support=96	85.4	0/162	85.0	65.8
n_support=144	84.5	0/162	85.0	65.8
risk scale=0.8	85.1	0/162	85.9	68.4
risk scale=0.9	85.1	0/162	85.9	67.4
risk scale=1.1	85.1	0/162	83.1	61.2
risk scale=1.2	85.1	0/162	78.7	50.8
SQ w=0.3333,0.3333,0.3334	85.1	0/162	85.0	65.8
SQ w=0.5,0.25,0.25	85.1	0/162	85.0	65.6
SQ w=0.2,0.4,0.4	85.1	0/162	85.0	65.8

TABLE IX

CUMULATIVE BUILD-UP ON THE HARD SET (%). FLIP: ADVISORY CHANGE UNDER MILD PERTURBATION (HARD SET)

Configuration	Unsafe	%	Over-c.	Severe	Flip
B0: YOLO only (road/depth priors)	1/30	3.3	59.7	49.0	13.0
B1: + HybridNets corridor	1/30	3.3	62.7	47.0	–
B2: + DA V2, single region	1/30	3.3	59.7	54.0	–
B3: + multi-region, scene quality	1/30	3.3	61.2	55.0	14.5
Full: + gates, fallback (proposed)	0/30	0.0	50.7	47.0	8.3

TABLE X

RELIABILITY STATES AGAINST BDD100K ANNOTATIONS (HARD SET, %)

State	n	False positive	Occl.	Trunc.	IoU
STABLE	437	11.9 [8.9, 15.1]	58.7	2.1	0.807
CHECK	67	23.9 [13.9, 34.2]	33.3	78.4	0.863
UNCERTAIN	1	100.0 [100.0, 100.0]	–	–	–

Pattern Recognition (CVPR), 2020, pp. 2636–2645.

[2] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth Anything V2," arXiv preprint arXiv:2406.09414, 2024.

[3] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[4] Q. M. Rahman, P. Corke, and N. Sünderhauf, "Run-time monitoring of machine learning for robotic perception: A survey of emerging trends," *IEEE Access*, vol. 9, pp. 20067–20075, 2021.

[5] D. Vu, B. Ngo, and H. Phan, "HybridNets: End-to-end perception network," arXiv preprint arXiv:2203.09035, 2022.

[6] Ultralytics, "Ultralytics YOLO documentation." [Online]. Available: https://docs.ultralytics.com/. Accessed: Jun. 13, 2026.

[7] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.

[8] D. Wu, M.-W. Liao, W.-T. Zhang, X.-G. Wang, X. Bai, W.-Q. Cheng, and W.-Y. Liu, "YOLOP: You only look once for panoptic driving perception," *Machine Intelligence Research*, vol. 19, no. 6, pp. 550–562, 2022.

[9] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, and V. Koltun, "Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 44, no. 3, pp. 1623–1637, 2022.

[10] L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, and H. Zhao, "Depth Anything: Unleashing the power of large-scale unlabeled data," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.

[11] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330.

[12] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?" in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[13] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[14] M. Poggi, F. Aleotti, F. Tosi, and S. Mattoccia, "On the uncertainty of self-supervised monocular depth estimation," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020.

[15] J. Hornauer and V. Belagiannis, "Gradient-based uncertainty for monocular depth estimation," arXiv preprint arXiv:2208.02005, 2022.

[16] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," arXiv preprint arXiv:2107.07511, 2021.

[17] D. Feng, A. Harakeh, S. L. Waslander, and K. Dietmayer, "A review and comparative study on probabilistic object detection in autonomous

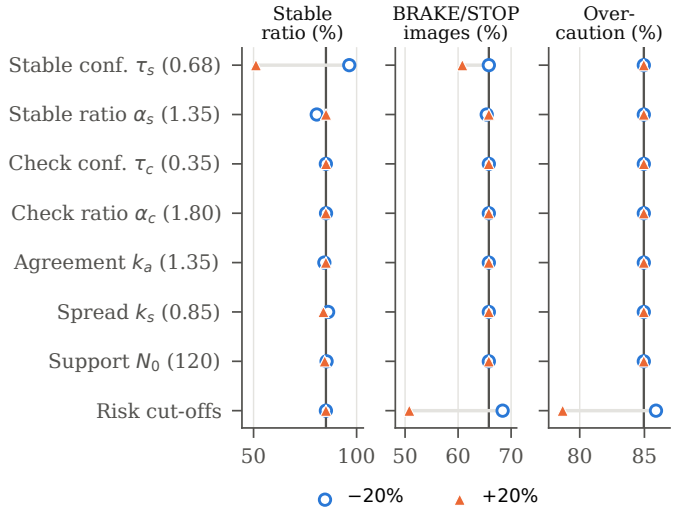


Fig. 5. Sensitivity on the dev subset: each fusion constant is changed by $\pm 20\%$ (the risk cut-offs are scaled together). Vertical rules mark the published constants.

driving," arXiv preprint arXiv:2011.10671, 2020.

[18] M. Bijelic, T. Gruber, F. Mannan, F. Kraus, W. Ritter, K. Dietmayer, and F. Heide, "Seeing through fog without seeing fog: Deep multimodal sensor fusion in unseen adverse weather," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020.

[19] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *J. American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.

[20] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[21] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.